# LIBRARY MANAGEMENT SYSTEM

## Python Midterm Project Report

**Team members:** Add the three names before submission.
**Course:** Programming Principles 2
**Submission:** October 18, 2026

## Project overview

This is a small console program for managing a library. It stores books, members, and borrowing records in SQLite. A short menu lets the user add, display, search, update, and cancel records, then view simple counts or save a text report.

## Database

| Table | Fields | Purpose |
|---|---|---|
| books | book_id, title, author, genre, available | Stores catalog details and availability |
| members | member_id, name, email, program | Stores student members and study programs |
| borrowings | borrowing_id, book_id, member_id, returned | Connects books with members and tracks return status |

The book_id and member_id fields in borrowings connect the three related tables. Sample member records are fictional.

# How the program works

## Classes

**Book** stores a book ID, title, author, genre, and availability. Its display method prints the details on one line. **Member** stores a member ID, name, email, and study program and prints the member details.

## Functions

| Function | Action |
|---|---|
| add_book / add_member | Save a new book or member |
| borrow_book / delete_record | Create or cancel a borrowing record |
| display_records / search_records | Show books, members, borrowing records, or matching results |
| update_book | Change a book title or author |
| show_statistics | Count borrowing records and active loans; find the most borrowed book |
| export_results | Write counts, borrowing statistics, and the list of loans to results.txt |

## Python concepts from class

The code uses basic variables, input, output, conditions, and a while loop for the menu. Functions handle each action. Book and Member are simple classes. SQLite stores the data, and a text file saves results. Regular expressions check member email addresses; random assigns sample borrowing records.

## Input checks

The program checks that required names are present, validates member email format, asks for number IDs, and prevents borrowing a book that is already out.

# Sample data and results

## Dataset

The included database has 50 varied book titles, 40 fictional student members, and 30 borrowing records. The catalog includes genre, and member records include a study program. Twelve loans are active and 18 are marked returned. The sample names and records are fictional.

## Statistics

| Calculation | Sample result |
|---|---|
| Total borrowing records | 30 |
| Books currently borrowed | 12 |
| Most borrowed book | Python Crash Course — 8 loans |

## Output from the running program

```
Choose an option: 9
Total borrowing records: 30
Books currently borrowed: 12
Most frequently borrowed book: Python Crash Course - 8 loans
```

## Team contributions

| Team member | Contribution |
|---|---|
| Member 1 - add name | Add the work completed and presentation part. |
| Member 2 - add name | Add the work completed and presentation part. |
| Member 3 - add name | Add the work completed and presentation part. |

Run with python3 main.py. Choose option 10 to create or refresh results.txt.